# Mathematics leads to the development of the world

Mathematics is a branch of science based on basic principles, logical reasoning and quantitative calculation (Knorr et al., 2024). Mathematics has been a foundation of human civilization as a universal language across diverse fields. Mathematicians say with some exaggeration, the history of mathematics mirrors the history of civilization. Huge advancements occur in society when mathematics has high valuation (Jayanthi, 2019). It allows humanity to reveal hidden patterns of the world. This is illustrated in the movie “The Imitation Game”. In this essay, the author will discuss whether and how mathematics has influenced societal development.

The movie “The Imitation Game” shows us that there are many unsolved problems in mathematics. Solving them has always been the most paramount factor in the development of mathematics, and furthermore the harmony in the world by mitigating human calamities.

One of them is that the number of results of a mathematical question could be vast or even infinite. It is of vital significance to improve your ideas and make them more efficient. In the movie, Turing and his team are told to break the Enigma code. The Enigma code updates every day, leaving only 24 hours for the team to work it out. Therefore, Efficiency becomes the key. At first, the team used to calculate all the possible results by human beings, which is extremely time-taking and inefficient. Turing designed and built a machine to decipher the cryptograph instead of human beings. But it is still not efficient enough to decipher the cryptograph within 24 hours. In a chat with a lady in the bar, Turing got inspired that the German will send some constant information in the first caught message every day, including some specific words like “weather”. This means that the information in the first message must refer to some constant words. Therefore, the machine does not need to calculate all the results. The machine can be more efficient by matching some specific words and cutting off many possibilities. Finally, within a few minutes, the machine worked out the cryptograph with only 1 specific word successfully matched. (Grossman, 2014)

Results for the Enigma are enormous. Therefore, the team used exhaustion to list all the possibilities, which is the most time-taking but universal skill. Then, Turing improved the idea by using pruning. With some known matching of certain words, cutting off many unnecessary possibilities, the total results the machine needs to process with get to be reduced. Pruning plays an important role in coding and decoding messages, having further impact on exploration on cryptography (Dong, W., Liu, J., Chen, L., 2024). It also highlights the importance of inspiration and effort. It is of great significance to find, catch and make good use of inspirations, which could be the key to success. Turing successfully catches and makes good use of the inspiration from the lady in the bar. That is the key to ultimate success. It took him 2 years to design and build the machine, which requires enormous amounts of effort. Inspiration and effort are the prerequisites for breaking the Enigma.

Aside from the technical problems, the external problems should not be ignored. There could be misunderstanding, distrust and biases towards mathematicians. In the movie, when Turing first got into the team, they did not understand or accept his measure at first, producing intense in the relationship. A bad relationship may get in the way of the research. Turing knows it is nearly impossible to build the machine by himself. Help is crucial for him. So, he tried to offer support to his teammates. For example, he brought them cups of coffee and told them a joke to strengthen the relationship among the teammates. As a reward, when Turing was in need, his teammates showed up and gave him strong support. This demonstrates the significance of a strong relationship. In addition, the government was not providing enough trust to Turing. For example, they did not give permission to produce the necessary component of the machine. Turing was dissatisfied with the government. So, he wrote to the prime minister. And it worked. Therefore, he became the leader of the team. Regardless of the social bias that being a homosexual is regarded as a crime, when someone makes a huge success, society tends to concentrate on the individual success, ignoring the discrimination towards him or her (Bien-Aimé, 2024). Apparently, social relationships and support are two key factors in breaking the Enigma, hence facilitate the development of mathematics.

With his talent, dedication and social support, Turing successfully broke the Enigma. Thus, the government is able to catch the essential information and message from the enemies to track them with corresponding measures, leading to the earlier ending of World War II. It is estimated that the war is shortened by over 2 years by breaking Enigma. More than 14 million people’s lives were saved. The whole civilization recovered and became organized again. Overcoming both technical and social challenges, Turing played the paramount role to mitigate human calamities by breaking the Enigma.

Apart from the contributions of math in helping Turing to mitigate human calamities, math had also led to great development in computer science, which will change the world, after the war. From the movie, Turing wanted to replace humans who calculate with pen and pencil to computing machine for decrypting Enigma. The replacement resulted in success in decrypting in a more intelligent way (“The Imitation Game”, 2014). It is obvious that the issues people met when using math would further encourage mathematicians or machine specialists to solve it by math, which shows that math inspires them to improve the word. Math further inspired other computer professionals to develop different core elements of computer science.

Take general-purpose computers as an example. During the 1940s, more and more computer scientists started to realize the universal Turing machine could be a mathematical model of general-purpose computer, which is an electronic machine with large storage containing both numbers and instruction (Daylight, 2015). The Electronic Numerical Integrator and Computer (ENIAC) was the first general-purpose computer, which was completed in 1945 and operated in 1946 by Mauchly and Eckert. It is a decimal computer, which could determine the artillery firing tables by complex calculation (Freiberger & Swanie, 2024). In 1946, the inventors of ENIAC also built a binary computer -- Electronic Discrete Variable Automatic Computer (EDVAC). Since the binary system only uses 0 or 1 to represent numbers, EDVAC can calculate addition, subtraction, multiplication, and division faster and more straightforwardly (Wilkes, 1956). Thanks to math, computer scientists could try different number systems in computers, which concluded by choosing binary system be the backbone of the general-purpose computer. Calculations could also be carried out regularly to build and operate the computer. We can clearly see that math is integral in the innovation of general-purpose computers.

In the movie, Turing mentioned an imitation game where a human judge interacts with both a machine and a human without knowing which one is a machine. In 1950, it advanced into the Turing Test, which could determine an intelligence level to the machine. A machine with a level passing the test was considered intelligent (Mijwil, 2015). A lot of computer scientists have delved into developing algorithms, artificial intelligence, and programming languages by math to enhance the behavior of computers, in order to pass the Turing test.

Alongside the design of the computer being completed, computer scientists began to build and apply algorithms in computers, again, based on the principle of math. Once applying the algorithm, the input data is computed through deterministic procedures every time the program is run (Hill, 2016). For example, the Dijkstra algorithm, which was developed in 1956, is one of the early algorithms in computer science. It finds the shortest distance between two nodes by calculating the distance from the starting node to an unvisited node. Once a shorter distance is calculated, the recorded shortest distance will be updated (GreeksForGreeks, 2024). There is no doubt that addition and comparison of numbers were applied in the algorithm, which shows that mathematical properties could be identified in the inspiration of algorithms in computer science.

With the successful cases of implementation in algorithms, computer practitioners started to investigate the fields of Artificial Intelligence (AI). The first AI system in the world was the logic theorist (Logic Theory-LT) program, which was invented by Neweell, Shaw and Simon in 1956. The AI system was for proving mathematical theories (Neweell & Simon, 1956). Again, it shows that mathematical problems were the primary focus in building AI systems. Consequently, math cannot be absent from the achievement in computer science.

In 1960, intermediate language algorithms and math were also used in the implementation of one of the first programming languages — Algol 60, which was completed by Dijkstra and others. The Algol 60 program is translated into an intermediate language, which is some tree structure data. The intermediate language will then be translated to the machine code read by the machine (as cited in Daylight, 2015). Since commands were interpreted into binary numbers which could be run on computers more simply and rapidly, the general-purpose computers became more flexible and powerful (Gorn and Manheimer 1956, p. 42, 58). Once again, the binary system was undoubtedly utilized in the development of elements of computer science.

The elements of computer science continued to expand and develop under the influence of math. In 1981, the first personal computer released by IBM started the popularity of personal computers (Mijwil, 2015). More programming languages have been developed for different uses, such as JavaScript for web development, and R for data analysis (Lang, 2024). Better algorithms have further strengthened different types of AI, including GPT language models (Clark, 2023). Thanks to math, computer science has become increasingly powerful in complex calculations and problem-solving, with its elements becoming pervasive and beneficial to society.

Mathematics has been a keystone of human advancement, which shapes our understanding of the world by influencing diverse fields. Among the various fields, computer science, biology and economics show the influence of mathematical development. Mathematical principles integrated into each of these disciplines to resolve complex problems, resulting in innovation and improvement.

Mathematics is fundamental to a various range of scientific and technological developments. Mathematical algorithms form the basics of programming and provide a foundation for data analysis (Jayanthi, 2019). For example, the software and hardware systems require discrete mathematics, which is an essential component. Moreover, mathematics allows human cognition model creation of artificial intelligence. In computer science, mathematics such as calculus, linear algebra and statistics are very important for producing neural networks and optimizing algorithms (Judijanto, 2024).

Mathematics integrated into computer science, resulting in many real-life applications and suggested a new paradigm for society. First, there's structural analysis for safety and integrity. In the architectural field, it helps to diagnose the safety and integrity of bridges and buildings. Computer science combined with mathematics has made it possible to build control systems for automation and robotics, which brought a huge impact on industry and the market economy (Jayanthi, 2019). In addition, mathematics is a cornerstone of information and communication technology, used in various techniques such as communication network design, traffic flow management and data compression for faster transmission (Judijanto, 2024).

Biology is another field where mathematical technology is required for profound understanding and further research on a biological complex system. The use of mathematical models allows biologists to simulate, analyze and predict biological processes. For example, in ecology, scientists can predict and analyze the growth and decay of population through mathematical modelling with applying calculus (Jayanthi, 2019).

In epidemiology, a mathematical model called the SIR mode is used. The most well-known example is the COVID-19 pandemic model (Khajanchi et al., 2021). It provided important information so that governments could adjust policies about outbreaks. It has played a major role during pandemic to protect public health.

Bioinformatics is an applied science, a combination of mathematics, computer science and biology. It interprets biological data using mathematical models and algorithms. It allows scientists to decode genomes, research genetic variation and develop personalized medicine (Jayanthi, 2019).

The role of mathematics in biology is diverse and important. If there was no integration between mathematics and biology, biologists wouldn't study to reveal the secrets of life's metabolisms and heredity. It confirms the mathematics is the cornerstone of biology that allows profound research.

In economics, all processes depend on the nature of number work and its interaction with reality (Judijanto el al., 2024). The mathematical components consist of economics such as mathematical modeling, statistical techniques and relevant logical reasoning for decision making (Yu, 2023). These mathematical elements are used in economics to analyze and predict variables in society such as consumer behavior, market fluctuations, and the distribution of resources. Moreover, when governments proposed economic policy, their evaluation is based on mathematical prediction (Yu, 2023). Furthermore, the user data and transcriptions are protected by cryptographic algorithms from external dangers (Jayanthi, 2019).

Mathematics integrated into various disciplines influencing societal advancement with lots of benefits. However, educational gap might be widened due to the requirement of understanding for more and more complex concepts (Jayanthi, 2019). It will build and limit the access to knowledge, and cause difficulty to mathematical approaches into science.

In conclusion, math can have a significant positive impact on industries and society, regardless of whether the current state of the world is good or bad. When we look at the big picture, it is easy to find that the influence of math in the world has never ceased. It seems that it will still last. Considering the circumstances, it is our firm belief that "Mathematics leads to the development of the world".

Reference

Bien-Aimé, S. (2024). "Hidden" No More: Newspapers' Framing of NASA Mathematician Katherine Johnson. Journal of Black Studies, 55(4), 310-328. https://doi-org.eproxy.lib.hku.hk/10.1177/00219347231225746

Clark, K. L. (1978). Negation as Failure. Springer EBooks, 293–322. https://doi.org/10.1007/978-1-4684-3384-5_11

Clark, S. (2023, December 21). The Era of AI: End of year AI recap. CMSWire.com. https://www.cmswire.com/digital-experience/the-era-of-ai-end-of-year-ai-recap/

Court, N. A. (1948). Mathematics in the History of Civilization. *The Mathematics Teacher*, *41*(3), 104–111. http://www.jstor.org/stable/27953289

Dong, W., Liu, J., Chen, L. et al. Implicit neural representation steganography by neuron pruning. Multimedia Systems 30, 266 (2024). https://doi-org.eproxy.lib.hku.hk/10.1007/s00530-024-01476-9

Edgar G. Daylight (2015) Towards a Historical Notion of 'Turing—the Father of Computer Science', History and Philosophy of Logic, 36:3, 205-228, DOI: 10.1080/01445340.2015.1082050

Freiberger & Swanie (2024). Encyclopedia Britannica. https://www.britannica.com/technology/von-Neumann-machine

geeksforgeeks. (2023, March 10). What is Dijkstra's Algorithm? | Introduction to Dijkstra's Shortest Path Algorithm. GeeksforGeeks. https://www.geeksforgeeks.org/introduction-to-dijkstras-shortest-path-algorithm/

Gorn, S. and Manheimer, W. 1956. The Electronic Brain and What it Can Do, Chicago: Science Research Associates, Inc.

Hill, R. K. (2016). What an Algorithm Is. Philosophy & Technology, 29(1), 35–59. https://doi.org/10.1007/s13347-014-0184-5

Jayanthi, R. (2019). Mathematics in Society Development - A Study. *Iconic Research and Engineering Journals*, *3*(3), 59–64. https://www.irejournals.com/formatedpaper/1701630.pdf

Jayanthi, R. (2019). Mathematics in Society Development - A Study. *Iconic Research and Engineering Journals*, *3*(3), 59–64. https://www.irejournals.com/formatedpaper/1701630.pdf

Judijanto, L., Manu, C., Sitopu, J., Mangelep, N., & Hardiansyah, A. (2024). The Impact of Mathematics in Science and Technology Development. *International Journal of Teaching and Learning*, *2*(2), 451–458. https://injotel.org/index.php/12/article/view/80

Knorr, W. R. , Fraser, . Craig G. , Berggren, . John L. , Folkerts, . Menso and Gray, . Jeremy John (2024, October 15). mathematics. Encyclopedia Britannica. https://www.britannica.com/science/mathematics

Lang, M. (2024, March 9). A History of Programming Languages. DEV Community. https://dev.to/mwlang/a-history-of-programming-languages-1i7f

Mijwil, M. M. (2015, April). History of Artificial Intelligence. ResearchGate; unknown. https://www.researchgate.net/publication/322234922_History_of_Artificial_Intelligence

Nanjundiah, V. (2003). Role of Mathematics in Biology. *Economic and Political Weekly*, *38*(35), 388–393. https://doi.org/10.2307/4413964

Newell, A., & Simon, H. (1956). The logic theory machine--A complex information processing system. IEEE Transactions on Information Theory, 2(3), 61–79. https://doi.org/10.1109/tit.1956.1056797

Schwarzman, Teddy et al. The Imitation Game. Hong Kong: EDKO Films Ltd., 2015. Film.

Subhas Khajanchi, Kankan Sarkar, Jayanta Mondal, Kottakkaran Sooppy Nisar, Sayed F. Abdelwahab (2021). Mathematical modeling of the COVID-19 pandemic with intervention strategies. Result in Physics, 25, https://doi.org/10.1016/j.rinp.2021.104285.

Turing, A. M. (1937). On Computable Numbers, with an Application to the Entscheidungsproblem. Proceedings of the London Mathematical Society, s2-42(1), 230–265. https://doi.org/10.1112/plms/s2-42.1.230

Tyldum, M. (Director). (2014). The Imitation Game [Film]. Black Bear Pictures & Bristol Automotive Orange Corp.

Wilkes, M. V. (1956). Automatic Digital Computers. New York: John Wiley & Sons.

Yu, S. (2023). Research on the Mathematical Application in Economics. *Advanced in Economics Management and Political Sciences*, *16*(1), 194–199. http://dx.doi.org/10.54254/2754-1169/16/20231006